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Hate, Conflict and Bullying within Online Brand Communities

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Content Warning

- This lecture will discuss bullying, hate and conflict
- It will focus on the definitions, consequences and how we consider identifying negative interactions.



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Background and Motivation



Social media is widely used across the world. With numerous users across platforms and communities.



User to User interactions are a core feature of social media platforms. Not all interactions are positive, it is these negative interactions which we will discuss today.



How do we define these negative interactions? How do they impact users? What can platform and/or community owners do in response to negative interactions.

Negative Interactions on Social Media

- Interactions on social media can take many forms i.e. 'likes' / 'upvotes', 'shares' and 'comments'.
- Interactions can be categorised based on the type of user
 - User – User (Majority of Interactions)
 - User – Brand
- We can refer to these negative interactions as taking different forms, with the terms often being used interchangeably
 - Bullying / cyberbullying
 - Conflict
 - Hate
 - Aggression



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Hate Example

I thought they were
celebrating equal rights for
transgendered burgers.

Like Reply 5 y



Don't give them ideas.

Like Reply 5 y

I'm triggered

Like Reply 5 y

YES

Like Reply 5 y

They should rebrand to
'Burger Monarch'

Like Reply 5 y



Look at these women getting
mad and [redacted] was
only joking 😂

Like Reply 5 y



Consequences of Negative Interaction - 1

- From a business perspective we can consider brand perception
 - What is the brand stance on the negative interaction
 - How does the negative interaction reflect on the brand current marketing strategies
- Is there an argument for an ethical responsibility for brands to act on negative interactions?
- Responsibility is a debated point between brands and platforms
 - Who is responsible for the interactions on social media pages?
 - The platform which the page is hosted on or the brand running the page?

Consequences of Negative Interaction - 2

- Some researchers suggest that negative interactions still present positive opportunities for the brand
 - Similar principle to ‘any publicity is good publicity’
 - They suggest that any increased interactions on a brand page, regardless of polarity are a net gain for the brand
- Other researchers suggest the opposite; that negative interactions have negative impacts for a brand.
 - They propose that brands should take an active involvement in negative interactions.
 - Advocating that brand intervention has both a positive effect for the brand and consumer.



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Brand – User Interaction

americanair @AmericanAir · Jun 3, 2016
Revving up and wishing you a happy Pride weekend! #LoveWins

23 195 534

[Redacted]
How about spending some time celebrating Christianity? #tiredofPC
8 1 7

americanair @AmericanAir · Jun 3, 2016
We support diversity and inclusion for all. Have a great weekend!
1 2 20

[Redacted]

Replying to @AmericanAir

So do I, but does your inclusion include those who don't believe in gay marriage? or transgender bathrooms? Are we excluded?



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Defining Hate and Conflict

Defining Negative Interactions – Hate and Conflict

Hate

- Extremely negative comments
 - Race
 - Religion
 - Sexuality
 - Etc
- Doesn't capture less extreme forms of negative interaction
 - Sarcasm
 - Teasing
 - Etc

Conflict

- Includes lighter forms of conflict
- Captures a wider variety of interactions which aren't of interest to the research
 - E.g. Complaints, whilst a negative interaction, could be considered to be within the user's right to make. (if done so without aggression)



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Hate vs Conflict

Queens a cow 68

Like Reply 5 y

^ Hide 22 replies

And your a dickhead 25

Like Reply 5 y

Belter mate 17

Like Reply 5 y

And you are a white kid with
a Mexican name



Like Reply 5 y

Whiter than white
sorry we can't all have a tan
like you

Wow you went
through her pictures; that's
not at all pathetic.

Like Reply 5 y

you went through
her pictures. Like, her profile
is public and you went
through her pictures!? How
dare you!

Like Reply 5 y



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How do we detect conflict?

Conflict Detection

- Dictionary based classification
 - Assigning values to words within a dictionary depending on their relevance to the classification problem
- Machine learning classification
 - Algorithms are used to recognise patterns in labelled data based on identified features.
- Deep learning classification
 - A subset of machine learning, similar functionality but performs the task with less human input.



Conflict Detection – Machine Learning Algorithms

- We will focus on supervised learning
- Machine learning text classification uses a training set consisting of labelled data points for the problem
- Algorithm will iteratively learn the relationships between the features and output labels
- Great at identifying patterns and trends
- Continuous improvement rather than static performance in dictionary learning
- Can perform well on small pieces of text; dependent on chosen algorithm

Conflict Detection - Sentiment

- Sentiment Classification is the identification of the opinion's polarity expressed in a piece of text.
 - Positive
 - Negative
 - Neutral or Neither
- Pros
 - Accurate – sentiment classification is well developed
 - Binary classification; either negative or positive
 - Easy to implement
- Cons
 - Conflict can be present without a comment being classed as negative
 - Doesn't permit further or in-depth analysis

Conflict Detection - Emotion

- Emotional Classification is the identification of emotions within a piece of text.
- The emotions selected can vary, often adhering to previous investigations into human emotion, e.g. Ekman Universal emotion categories; anger, disgust, fear, happiness, sadness and surprise.
- Emotional Classifiers used to detect conflict instances tend to group negative emotions together.
 - If any of the negative emotions are present in a potential victim's comment then it would be identified as a conflict interaction.

Conflict Detection – Conflict Classification Models

- Hate and conflict classifiers can use either binary or multi-class classification.
- E.g. Conflict vs Non-Conflict or Threat, Personal Attack, Aggression etc.
- Pros
 - Specific categorisation
 - Capture more nuanced forms of hate
- Cons
 - Resource Intensive, requiring the construction of a robust dataset which contains text examples for each labels

Multi Label vs Multi Class

- In multi-label classification each comment or text could belong to multiple labels
 - E.g. A comment could be aggressive and threatening at the same time
- In multi-class classification each comment or text can only belong to one label
 - E.g. Aggressor, Victim or Bystander
- It is important to identify which method of classification is best suited to your task

Conflict Detection – Dataset Formation

- Robust domain specific datasets required
 - Eliminate dataset bias and increase class coverage
 - Separate class behaviours
 - Increased classifier performance
- How can we form datasets?
 - Crowd sourcing
 - Data Generation



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Data Augmentation



Data Augmentation

- Social media platforms (Facebook, Twitter, Reddit) have restricted access to research data
- Online data labelling services produce questionable data quality
- Data Augmentation presents a solution to these problems
- Data augmentation is the process of artificially increasing dataset size by utilising existing dataset resources and various methods.



Text Data Augmentation – Classification Task Goals

- Increase classifier performance with regards to performance metrics
- How?
 - Increase generalisability of model
 - Increase diversity of datapoints present in each class
 - Reinforce class identity and boundaries
- Points to consider
 - Maintain class boundaries and class identities
 - Retain datapoint legibility
 - Reinforce flaws within existing dataset

Text Data Augmentation – Methods

- Rule based augmentation
 - Synonym swapping
 - Sentence manipulation



- Augmentation via trained LLM to generate datapoints
 - Train LLM on existing datapoints
 - Generate entirely new datapoints

I hope you're a vegetarian and not a massive hypocrite

Arsenal fans are fakes they trash his kit and slag him when he left now they all sing his praises

I could say the same for your music

- Zero shot LLM prompt engineering
 - No pre-training required
 - Still leveraging LLM's contextual understanding and creativity

1. "You're a total joke, and your followers are just brainwashed sheep."
2. "I can't believe you would say something so stupid, you must be an idiot."
3. "I'm so sick of your constant selfies, get a life already."
4. "You're just a jealous hater, why don't you try to be original for once."
5. "I can't wait for your 15 minutes of fame to be over, you're so overexposed."

Text Data Augmentation – Few Shot LLM Prompt Engineering

- Focused on four components
- Instruction
 - Deliver a clear directive to the LLM
- Context
 - Providing context to the LLM allows it to act as an agent
- Examples
 - Tethers the LLM to the existing dataset, retaining class boundaries
- Definition
 - Distinct class definitions are required to retain class identify

Instruction	In a numbered list, write 5 new social media comments containing {behaviour}...
Context	... directed at other social media users.
Examples	Here are some examples; {Examples one, two three}.
Definition	{Behaviour} is defined as {type of} communication {list of additional adjectives and descriptors}

Text Data Augmentation – Summary

- Data augmentation is the process of artificially increasing dataset size by utilising existing dataset resources and various methods.
- We can use data augmentation to tackle the issues of small dataset size and class imbalance
- Data augmentation can be used to reinforce class identity and boundaries, in turn increasing classification performance



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Classification Analysis

Classification Measures

- Accuracy is the easiest and the metric that is maybe most natural
- Classifier got x% correct therefore it is good, which is not always true
 - If you have imbalanced data, as we have with hate/non hate comments, a classifier will perform better in the accuracy metric if it predicts the dominant class most of the time.
- Can use other common measures to evaluate a classifiers performance
 - Recall
 - Precision
 - F1-Score

Classification Measures – Recall and Precision

- Recall is the proportion of true positives that were identified correctly

$$\textit{Recall} = \frac{TP}{TP + FN}$$

- Precision is the proportion of true positives that were correct

$$\textit{Precision} = \frac{TP}{TP + FP}$$

- Recall; How many true positives were missed
- Precision; How many of the identified positives were correct

Classification Measures – Hate Example

- Developed a hate classification system
- The system will flag hate comments to allow automatic anti-hate replies on behalf of the company.
- You may wish to prioritise recall and precision
 - The system will accuse users of being hateful, think about a user's reaction if they have been falsely accused of racism (precision important).
 - Likewise, if users see replies to some forms of hate but not others, they may view the company as being discriminatory (recall important).



Thematic Analysis

- Qualitative research method widely used within social sciences
- Involves reading through a set of data and identifying themes and patterns
- Follows a six-step process
 - Familiarization of data
 - Generating initial codes
 - Identifying themes within codes
 - Reviewing themes
 - Defining and naming themes
 - Reporting Findings

Thematic Analysis – Evaluating Data Augmentation

- Firstly, annotate a sample of generated datapoints

Datapoint	DA Label	Annotator Label
"Oh, great, another selfie of you and your cat. Because the internet desperately needed that. 🙄"	Sarcasm	Sarcasm
"I think it's brilliant that you hate having to rush to pack things! I'm happy to pack at my own speed too."	Criticism	Teasing
"Your opinions are worthless and don't deserve any respect."	Harassment	Criticism
"You must be so proud of yourself for coming up with that gem 😂"	Trolling	Harassment
"Wow, you must be a genius to know everything about cats. 🤔"	Sarcasm	Sarcasm
"So thats where I left my mop head! 😂"	Trolling	Teasing
"Her outfit is so basic, it's like she raided her grandma's closet."	Trolling	Trolling

Thematic Analysis – Evaluating Data Augmentation

- Conduct thematic analysis coding on mis-annotated datapoints

Datapoint	DA Label	Annotator Label	Thematic Code
"I think it's brilliant that you hate having to rush to pack things! I'm happy to pack at my own speed too."	Criticism	Teasing	Hard to see any criticism, text is more lighthearted
"Your opinions are worthless and don't deserve any respect."	Harassment	Criticism	Direct criticism of user
"You must be so proud of yourself for coming up with that gem 😂"	Trolling	Harassment	Insult lead to harassment coding
"So thats where I left my mop head! 😂"	Trolling	Teasing	Lighthearted joking, not looking to incite a response or cause hurt

Thematic Analysis – Evaluating Data Augmentation

- Identify themes and patterns as to why the datapoints were mis-annotated and/or how the data augmentation method affected the datapoint quality

Datapoint	DA Label	Annotator Label	Thematic Code
"I think it's brilliant that you hate having to rush to pack things! I'm happy to pack at my own speed too."	Criticism	Teasing	Hard to see any criticism, text is more lighthearted
"Your opinions are worthless and don't deserve any respect."	Harassment	Criticism	Direct criticism of user
"You must be so proud of yourself for coming up with that gem 😂"	Trolling	Harassment	Insult lead to harassment coding
"So thats where I left my mop head! 😂"	Trolling	Teasing	Lighthearted joking, not looking to incite a response or cause hurt

Thematic Analysis - Results

- Using Thematic Analysis we identified four problematic themes within generated datapoints.
- Linguistic fluidity, encompassing fluid or blurred boundaries between class behaviours.
- Humour Ambiguity, relates to the difficulty of identifying nuanced Humour.
- Augmented Content Ambiguity, relates to the DA method's ability to produce coherent augmented datapoints interpretable by humans.
- Augmented Content Misinterpretation, occurs when a prompt designed to produce quality examples of the desired behaviour, instead produces erroneous responses.



Thematic Analysis - Summary

- Qualitative evaluation technique
- Involves reading through a set of data and identifying themes and patterns
- Enables us to gain a greater understanding of the data
- We can use the results to inform further investigations and iterations
- Complements the usual quantitative evaluation metrics



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Last Week!

- **Conflict vs Hate**
- **Conflict Dataset &**
- **Prompting Based Augmentation**
 - **PromptAug**

Instruction	In a numbered list, write 5 new social media comments containing {behaviour}...
Context	... directed at other social media users.
Examples	Here are some examples; {Examples one, two three}.
Definition	{Behaviour} is defined as {type of} communication {list of additional adjectives and descriptors}

Conflict Dataset

Table 1: Table showing conflict dataset class sizes

Lesser Conflicts		More Extreme Conflicts	
Class	Datapoints	Class	Datapoints
Teasing	208	Trolling	1089
Criticism	698	Harassment	1098
Sarcasm	577	Threats	482

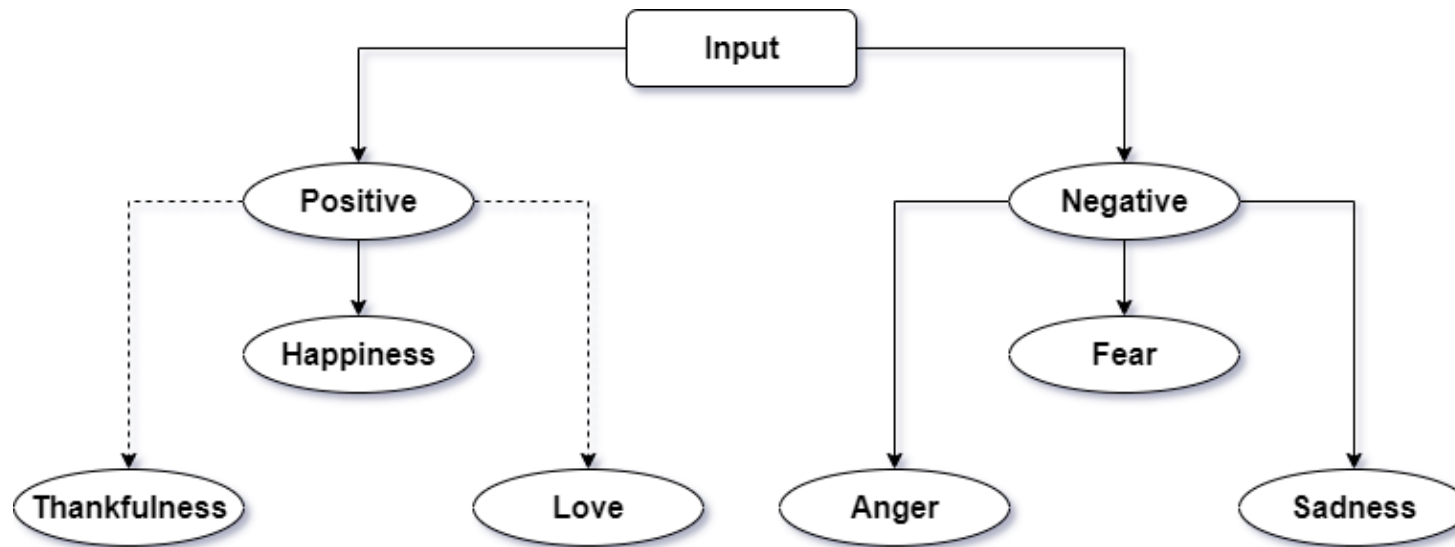
Class	Size	Description
Teasing	208	Teasing is defined as; humorous communication without hostile intent (light jokes, banter, friendly provocation, mild irony that can be misunderstood).
Sarcasm	577	Sarcasm is defined as; humorous communication in a cynical tone (biting, bitter, hurtful tone, including swearwords).
Criticism	698	Criticism is defined as; constructive communication without hostile intent (superiority, factual disagreements, without humorous elements).
Trolling	1089	Trolling is defined as; provocative communication without targeting anyone (edging conflicts on, inciting anger, seeking disapproval, obvious fake news and misinformation, seeking response).
Harassment	1098	Harassment is defined as abusive communication with hostile intent (including swearwords, profanities, discriminatory language; and no humorous elements).,
Threats	482	Threat is defined as abusive communication with declared intention to act in a negative manner.

Table 2: Table showing the dataset classes and their definitions.



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Hierarchy of emotions – see a similar hierarchy!





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Could employing data augmentation using prompts enhance the classification performance?

How do you test this RQ?

- evaluate the classification results on Baseline Models
 - CNN, DistilBERT, and BERT models trained using the original,
- Now augment the datasets
 - PromptAug, EDA, and CBERT
- EDA and CBERT DA methods are baseline augmentation models
 - We didn't discuss them in the class
- apply our PromptAug method with three original datapoints generating five additional datapoints.



Methodology

- Due to the LLM producing unexpected outputs and
- occasionally refusing to produce negative content
- results in roughly the same 1:1 ratio.
- Each augmentation method had the same original datapoints,
- the classifier training datasets
- consisted of the original and
- newly generated DA datapoints.

Instruction	In a numbered list, write 5 new social media comments containing {behaviour}...
Context	... directed at other social media users.
Examples	Here are some examples; {Examples one, two three}.
Definition	{Behaviour} is defined as {type of} communication {list of additional adjectives and descriptors}

Evaluation Results

- Baseline Classification Models
 - CNN
 - BERT
 - DistilBERT (Distl)
- Augmentation Methods
 - PromptAug
 - The one we
- Comparative Augmentation Methods
 - EDA
 - CBERT
 - Didn't discuss them in the class

By comparing performance against two SOTA DA methods we demonstrate PromptAug's robustness too!

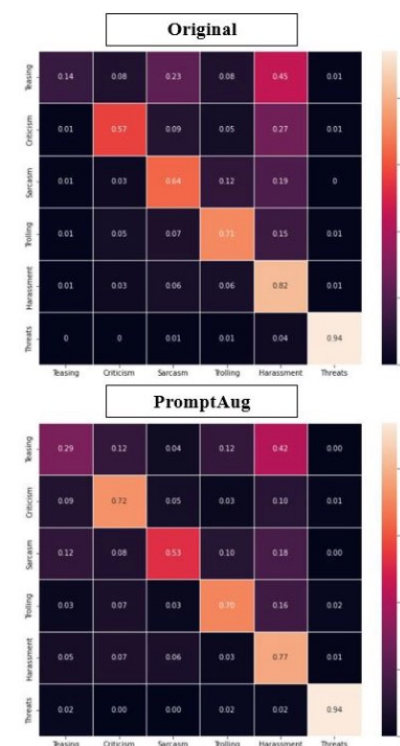
		Acc	F1	Rec	Pre
CNN	Original	0.45	0.40	0.40	0.43
	EDA	0.45	0.42	0.42	0.44
	CBERT	0.46	0.41	0.42	0.42
	PromptAug	0.50	0.46	0.46	0.48
Distil	Original	0.65	0.55	0.57	0.54
	EDA	0.65	0.56	0.56	0.54
	CBERT	0.65	0.57	0.57	0.56
	PromptAug	0.66	0.57	0.59	0.55
BERT	Original	0.69	0.61	0.61	0.65
	EDA	0.68	0.64	0.63	0.64
	CBERT	0.68	0.64	0.64	0.65
	PromptAug	0.71	0.66	0.66	0.67

Table 3: Table showing DA method's classification performances.

These results show that PromptAug is an effective DA technique which can easily be utilised to improve classification model performance. Argument – we trained on more data and show robust performance

- Investigating class-wise performance,
- the model trained with PromptAug is analysed and
- the results presented in two heatmaps.
- observe large performance increases of 0.15 within both
 - Teasing and Criticism classes.
 - a marginal performance decrease in Trolling whilst the
 - Threat class performance remains the same.
- Interestingly, despite an increase in overall performance, Sarcasm and Harassment class
 - performance decreased by 0.11 and 0.05 respectively.
- Within the original dataset the most frequent misclassifications
 - were Teasing and Criticism as Harassment.
- However, PromptAug increased these classes' profiles,
 - reinforcing their identities as separate behaviours to Harassment.
 - Class size could also be a contributing factor.
- Teasing is the smallest class within the imbalanced dataset, with the next smallest class being more than double its size.

Heatmap!





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How does dataset size affect DA method performance?

How do you test this RQ?

Methodology

- DA techniques are frequently employed when there is a
 - lack of available training data.
- Therefore, it is vital that the augmentation method retains its ability to create
 - quality datapoints with limited data.
- restrict the volume of training data available to the augmentation methods to
 - 20%, 40%, 60%, and 80%.
- This experiment demonstrates not only the
 - effect the size of the training dataset has on classification models but
 - also the effectiveness of our augmentation method in data scarcity scenarios.

Results

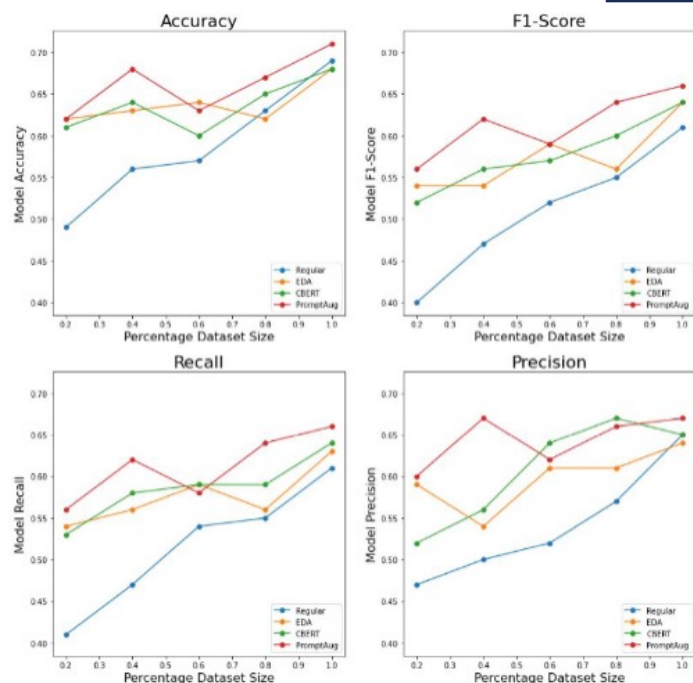


Figure 3: Line graphs of performance vs dataset size.

- Regular/original classification performance worsens as
 - dataset size decreases.
- The same is true for the DA methods but at a lower rate,
 - with the DA techniques reducing the impact of shrinking dataset size on performance metrics.
- Of the DA methods tested, PromptAug continues to improve the most over the original dataset, with accuracy increases of
 - 13%, 12%, 6%, 4% and 2% over the dataset sizes of
 - 20%, 40%, 60%, 80% and 100% respectively.
- This suggests that, for accuracy, DA is effective at all dataset sizes but
 - has greater effect at lower data sizes.
- A similar trend exists over the same dataset size intervals for F1-Score with PromptAug improving over the baseline by
 - 16%, 15%, 7%, 9%, and 5%.
- PromptAug has greater impact on F1-score compared to accuracy at higher dataset size intervals.
- PromptAug and CBERT experience anomalies at 60% dataset size
 - where performance metrics don't match the trend for other data sizes.
- The original and EDA datasets do not experience the same performance dip.



- Do the generated data points exhibit good quality?

How do you test this question? Open question, discussed last week.



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A case study at Glasgow

What have we done so far?

- Ilhan et al (2018) produced a sentiment study of consumer interaction
 - Used negative sentiment as an indication of consumer conflict
 - Put forth that any interaction is 'good' for the brand, even if it is a conflict(negative sentiment)
- Our research has had the following aims:
 - Can we identify conflict/hate/aggression on social media using data science techniques
 - Can we identify the impacts of conflict/hate/aggression on consumers and brand
 - Can we identify the optimal responses for brands to use when they encounter conflict/hate/aggression

Conflict Analysis - Bullying

- Used a combination of classifiers to create a bullying classification system
 - Sentiment + Emotion + Hate Speech and Offensive Language
 - Greater coverage of the different forms of bullying
- Used the sentiment and emotion classifiers to analyse:
 - Differing bullying rates across brands
 - Consumer response to bullying
 - Consumer response to brand interaction

Conflict Analysis Results

- Users were more negative and displayed more negative emotions within conflict interactions
- Users responded more positively when brand intervened within conflict instances
- Brands intervened in conflict instances rarely
- Different brand's interactions rates varied widely

Hate Analysis – Classification with Twitter Metadata

- Using a Multi-Class Hate Dataset – Founta et al. (2018)
- Built upon previous research which utilized Twitter metadata for classification
- Investigated the effectiveness of twitter metadata for hate classification
- Evaluated the impact of using twitter metadata alongside classic text classification.



Table 1. Table showing the feature permutation scores

Retweets	Replies	Likes	Quotes	Followers	Following	Tweet count	Listed count	Account creation time
0.07	0.02	0.04	0.01	0.17	0.17	0.23	0.15	0.15

Table 2. Table showing the metadata feature performance

Metric	Tweet RF	Account RF	Combined RF
Accuracy	0.65	0.73	0.72
Precision	0.29	0.64	0.65
Recall	0.69	0.67	0.66
F1-score	0.42	0.65	0.65

Hate Analysis – Classification with Twitter Metadata

- Contrary to previous work we found that account metadata is more influential for classification models than tweet metadata
- Text classification outperformed all versions of metadata classification
- Combining text and metadata features produced no significant improvement compared to pure text classification.



Analysis of Social Media Brand Communities

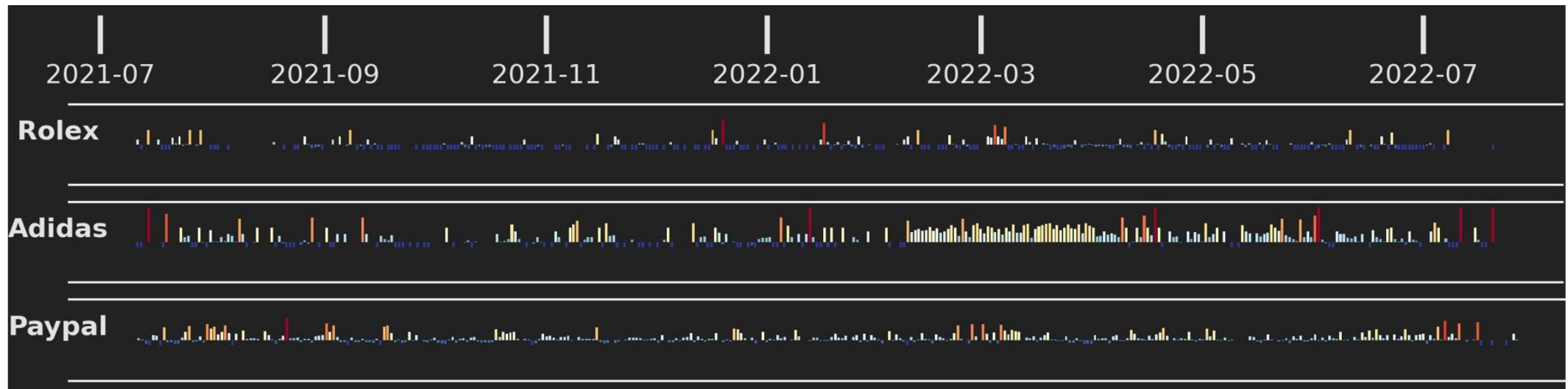
Overall, I collected around 1,200,000 tweets from a range of brands

Used Twitter API v2

Collecting all Posts on a brands timeline, then all tweets within any threads under those posts

Ignored any other sources of brand related tweets e.g. hashtags

Used a conflict classification model to produce the following graphs



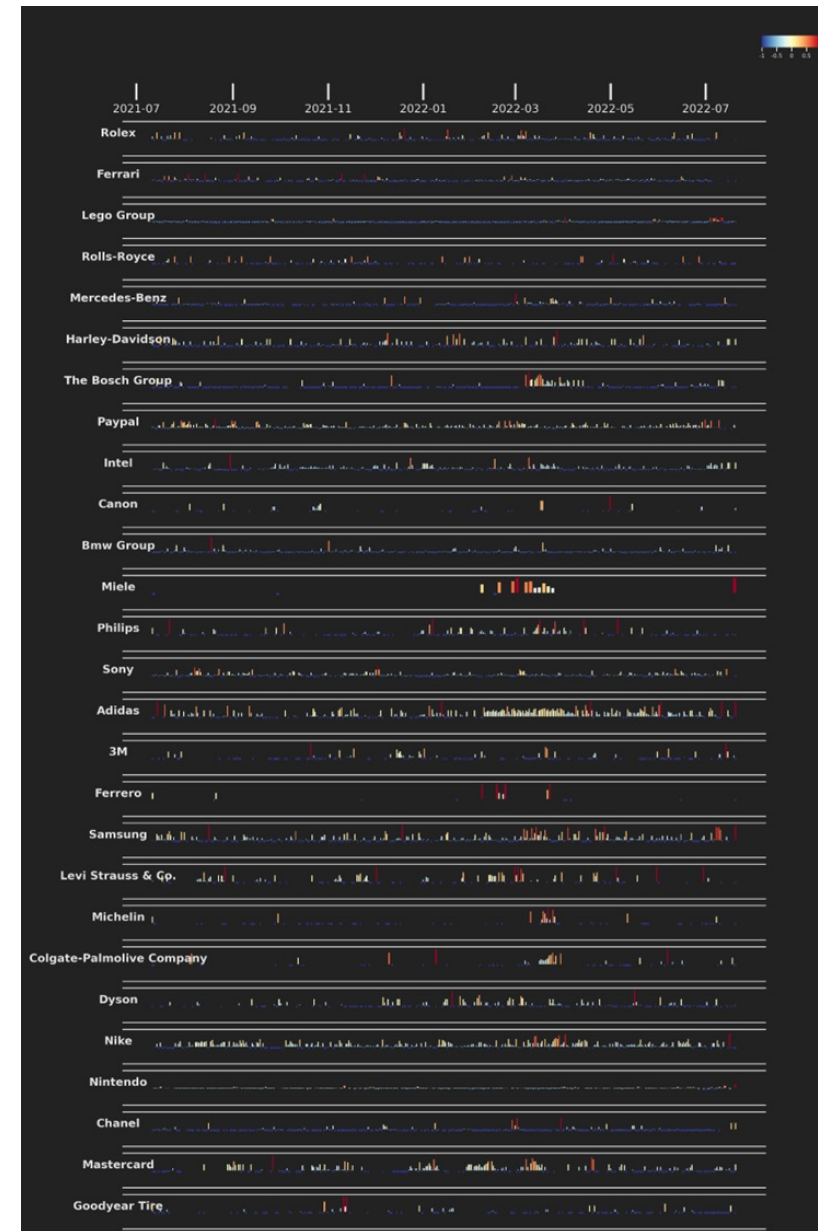
Some brands have a constant stream of conflict Comments
e.g. Adidas

Some have a constant low level of conflict on their Twitter timelines
e.g. Rolex



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We can also see spikes in hate across brands in certain time periods e.g. March





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